

Questions

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Q1 - 2024 (04 Apr Shift 1)

Two wavelengths λ_1 and λ_2 are used in Young's double slit experiment. $\lambda_1 = 450 \text{ nm}$ and $\lambda_2 = 650 \text{ nm}$. The minimum order of fringe produced by λ_2 which overlaps with the fringe produced by λ_1 is n . The value of n is _____.

Q2 - 2024 (04 Apr Shift 2)

The width of one of the two slits in a Young's double slit experiment is 4 times that of the other slit. The ratio of the maximum of the minimum intensity in the interference pattern is:

- (1) 1 : 1
- (2) 4 : 1
- (3) 9 : 1
- (4) 16 : 1

Q3 - 2024 (05 Apr Shift 1)

Light emerges out of a convex lens when a source of light kept at its focus. The shape of wavefront of the light is :

- (1) both spherical and cylindrical
- (2) plane
- (3) spherical
- (4) cylindrical

Q4 - 2024 (05 Apr Shift 1)

In Young's double slit experiment, carried out with light of wavelength $5000 \text{ \AA}$, the distance between the slits is 0.3 mm and the screen is at 200 cm from the slits. The central maximum is at $x = 0 \text{ cm}$. The value of x for third maxima is _____ mm.

Q5 - 2024 (05 Apr Shift 2)

Questions

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In a single slit experiment, a parallel beam of green light of wavelength 550 nm passes through a slit of width 0.20 mm. The transmitted light is collected on a screen 100 cm away. The distance of first order minima from the central maximum will be $x \times 10^{-5}$ m. The value of x is :

Q6 - 2024 (06 Apr Shift 2)

Two coherent monochromatic light beams of intensities I and $4I$ are superimposed. The difference between maximum and minimum possible intensities in the resulting beam is xI . The value of x is \$_____.

Q7 - 2024 (08 Apr Shift 1)

A parallel beam of monochromatic light of wavelength 600 nm passes through single slit of 0.4 mm width.

Angular divergence corresponding to second order minima would be _____ $\times 10^{-3}$ rad.

Q8 - 2024 (08 Apr Shift 2)

Two slits are 1 mm apart and the screen is located 1 m away from the slits. A light of wavelength 500 nm is used. The width of each slit to obtain 10 maxima of the double slit pattern within the central maximum of the single slit pattern is _____ $\times 10^{-4}$ m

Q9 - 2024 (09 Apr Shift 1)

In a Young's double slit experiment, the intensity at a point is $\left(\frac{1}{4}\right)^{\text{th}}$ of the maximum intensity, the minimum distance of the point from the central maximum is _____ μm .

(Given : $\lambda = 600$ nm, $d = 1.0$ mm, $D = 1.0$ m)

Q10 - 2024 (09 Apr Shift 2)

Monochromatic light of wavelength 500 nm is used in Young's double slit experiment. An interference pattern is obtained on a screen. When one of the slits is covered with a very thin glass plate (refractive index = 1.5), the central maximum is shifted to a position previously occupied by the 4th bright fringe. The thickness of the glass-plate is _____ μm .

Questions

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Answer Key

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Q1 (9) mathongo Q2 (3) mathongo Q3 (2) mathongo Q4 (10) mathongo

Q5 (275) mathongo Q6 (8) mathongo Q7 (6) mathongo Q8 (2) mathongo

Q9 (200) mathongo Q10 (4) mathongo mathongo mathongo mathongo

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Solutions

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Q1

$$n_2 \lambda_2 = n_1 \lambda_1$$

$$\frac{n_2}{n_1} = \frac{\lambda_1}{\lambda_2} = \frac{450}{650} = \frac{9}{13}$$

$$n_2 = 9$$

Q2

Since, Intensity $\propto$ width of slit (ω)

so, $I_1 = I, I_2 = 4I$

$$I_{\min} = \left(\sqrt{I_1} - \sqrt{I_2} \right)^2 = I$$

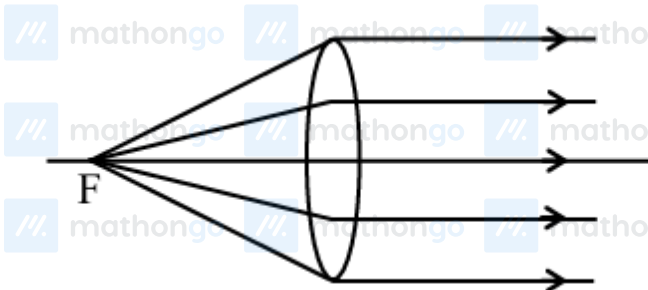
$$I_{\max} = \left(\sqrt{I_1} + \sqrt{I_2} \right)^2 = 9I$$

$$\frac{I_{\max}}{I_{\min}} = \frac{9I}{I} = \frac{9}{1}$$

Q3

Light emerges parallel

$\therefore$ planor wavefront



Q4

$$\beta = \frac{\lambda D}{d} = \frac{5 \times 10^{-7} \times 2}{3 \times 10^{-4}} = \frac{10 \times 10^{-3}}{3} \text{ m}$$

For 3rd maxima $y_3 = 3\beta = 10 \times 10^{-3} \text{ m} = 10 \text{ mm}$

Q5

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Solutions

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$$y = \frac{\lambda D}{d} = \frac{550 \times 10^{-9} \times 100 \times 10^{-2}}{0.2 \times 10^{-3}} = 275$$

Q6

$$I_{\max} = (\sqrt{I} + \sqrt{4I})^2 = 9I$$

$$I_{\min} = (\sqrt{4I} - \sqrt{I})^2 = I$$

$$\therefore I_{\max} - I_{\min} = 8I$$

Q7

$$\sin \theta \simeq \theta \simeq \frac{2\lambda}{b}$$

$$= \frac{2 \times 600 \times 10^{-9}}{4 \times 10^{-4}} = 3 \times 10^{-3} \text{ rad}$$

$$\text{Total divergence} = (3 + 3) \times 10^{-3} = 6 \times 10^{-3} \text{ rad}$$

Q8

$$d = 1 \text{ mm}, D = 1 \text{ m}, \lambda = 500 \text{ nm}$$

$$10 \left(\frac{\lambda D}{d} \right) = \frac{2\lambda D}{a}$$

$$a = \frac{d}{5}$$

$$= \frac{10 \times 10^{-4} \text{ m}}{5}$$

$$= 2 \times 10^{-4}$$

Q9

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Solutions

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$$I = I_0 \cos^2 \left(\frac{\Delta \phi}{2} \right)$$

$$\frac{I_0}{4} = \cos^2 \left(\frac{\Delta \phi}{2} \right)$$

$$\Delta \phi = \frac{2\pi}{3}$$

$$\frac{2\pi}{\lambda} \left(\frac{y d}{D} \right) = \frac{2\pi}{3}$$

$$y = \frac{\lambda D}{3 d} = \frac{600 \times 10^{-9} \times 1}{3 \times 10^{-3}} = 2 \times 10^{-4} \text{ m}$$

Q10

$$(\mu - 1)t = n\lambda$$

$$(1.5 - 1)t = 4 \times 500 \times 10^{-9} \text{ m}$$

$$t = 4000 \times 10^{-9} \text{ m}$$

$$t = 4 \mu\text{m}$$

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